



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: MLA0408

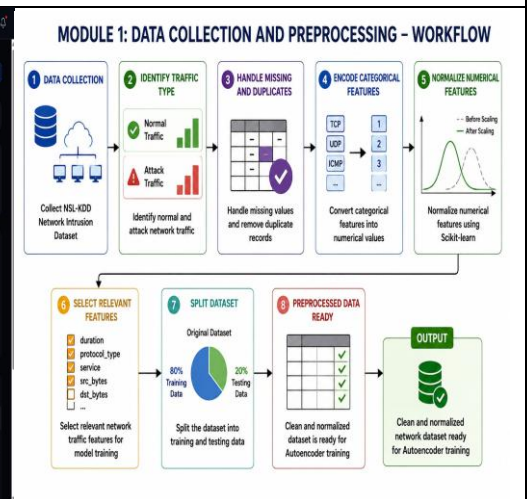
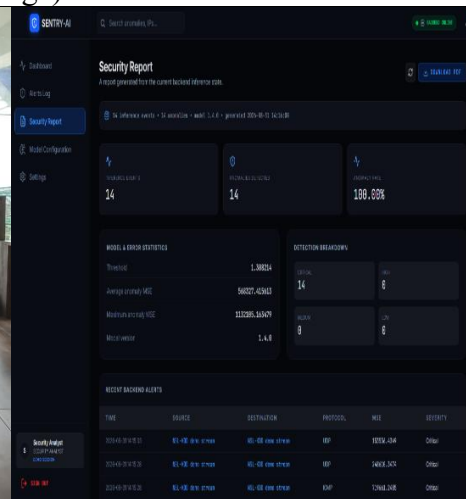
Slot: C

Course Name: Deep Learning

Course Faculty: Dr. D. SUDHAGAR

Project Title: Autoencoder-Based Deep Learning Framework for Network Intrusion and Anomaly Detection.

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

Autoencoder-Based Deep Learning Framework for Network Intrusion and Anomaly Detection is a cybersecurity system that uses an **autoencoder deep learning model** to detect unusual network activities.

The model learns normal network traffic patterns and identifies anomalies based on **reconstruction error**. The system includes data preprocessing, model training, anomaly detection, and performance evaluation using accuracy, precision, recall, and F1-score. It helps detect both known and unknown network attacks efficiently.

Student Signature

Guide Signature